

Programme Name : B.Tech

Semester : II

Course Name : DSA

Time : 03 hrs

Course Code : CSEG 1043

Max. Marks: 100

Nos. of page(s) : 2

Instructions: Answer all the questions.

SECTION A

S. No.		Marks	CO
Q 1	Explain the advantages of a circular queue over a linear queue. Illustrate your answer with the help of a suitable diagram.	4	2
Q 2	What is a dangling pointer in C? Explain how it is created and how it can be avoided. Illustrate your answer with a suitable example.	4	1
Q 3	Write a C program to perform binary search on a sorted array and display whether the element is found or not.	4	4
Q 4	Determine the number of new nodes required to convert a perfect binary tree of height h into an extended binary tree. Show all steps and justify your answer.	4	3
Q 5	A hash table has a high load factor and is experiencing frequent collisions. Which collision resolution technique would perform better and why?	4	3

SECTION B

Q 6	a. Write a C function to delete an element from a min-priority queue implemented using an array. b. Suggest the most appropriate sorting technique for a nearly sorted dataset. Justify your choice and explain why it is preferred over other sorting algorithms.	5+5	3, 4
Q 7	Write C functions to implement push and pop operations in a stack using an array. Also convert the below expression from infix to postfix using stack. $A+(B*C-(D/E^F)*G)*H$	10	2
Q 8	Apply Quick Sort to the following array using the divide and conquer approach: 38, 27, 43, 3, 9, 82, 10 Show all steps involved in partitioning and recursive division of the array until it is completely sorted. Also, state the time complexity of Quick Sort in best, average, and worst cases.	10	4
Q 9	Write a C program to implement a Generalized Linked List with operations for insertion at the end, deletion from the beginning, and display of elements	10	2

OR

Write a C program to create a singly linked list and reverse the list in-place (without using any extra array or additional data structure). Display the elements of the list before and after reversal.

SECTION C

Q 10

- a. Insert the keys 15, 28, 19, 20, 33, 12, 17, 26 into an initially empty hash table of size 13 using linear probing. Show all steps of insertion, clearly indicating each probe sequence and collision handling. Also, construct the final hash table.
- Further, give a set of any five keys such that insertion using linear probing leads to worst-case time complexity, and briefly justify your answer.
- b. An undirected weighted graph has 7 vertices (A–G) with the following edges and corresponding weights: (A–B, 2), (A–D, 6), (B–C, 3), (B–D, 8), (B–E, 5), (C–E, 7), (D–E, 9), (E–F, 4), (E–G, 2), (F–G, 1). Construct the graph. Apply Prim's Algorithm to find the Minimum Spanning Tree (MST). Show all steps. Find the total cost of the MST.

10/10 3

Q 11

Write a C program to implement a Binary Search Tree (BST) with functions for insertion of nodes, finding the height of the tree, counting the total number of leaf nodes, and performing tree traversals (inorder, preorder, and postorder). Also, construct a binary tree from the following given traversal orders:

Inorder : D B E A F C H G I J
Postorder : D E B F H I J G C A

OR

Using the given data 40, 20, 30, 70, 80, 50, 25, 28, construct the following trees by inserting the elements in the given order:

(i) AVL Tree

(ii) B-Tree of order 3 ($m = 3$)

Show the construction step-by-step and clearly indicate all rotations and node splits, wherever applicable. Also provide the level-order, inorder, postorder, and preorder traversal of the constructed AVL tree.

20 3